

Data Vortex 2026 - Round 2 Technical Report

1. Problem Definition

The Social Engine requires a semantic layer capable of interpreting short human-language posts.

This project builds two classifiers:

```
post_text -> sentiment_label  
post_text -> topic_category
```

The sentiment model predicts Positive, Negative, or Neutral. The topic model predicts the category of the post.

2. Dataset Overview

The dataset contains 9,000 records and four source columns:

```
text_id  
post_text  
sentiment_label  
topic_category
```

The inspection found no missing values, blank text values, duplicate complete rows, or duplicate text IDs.

There are 7,900 unique text values and 987 repeated text values. Repeated texts have no conflicting labels.

The sentiment labels are balanced with 3,000 records per class. The topic labels are strongly imbalanced.

3. Preprocessing Pipeline

The preprocessing pipeline:

1. Decodes HTML entities.
2. Decodes escaped Unicode sequences where possible.
3. Replaces URLs with **URLTOKEN**.
4. Replaces mentions with **USERTOKEN**.
5. Normalizes whitespace.
6. Converts text to lowercase.
7. Creates TF-IDF unigram and bigram features.

TF-IDF used minimum document frequency 2, sublinear term frequency, and a maximum of 100,000 features.

4. Model Selection

Two models were compared:

- Logistic Regression.
- Linear Support Vector Machine.

Both used the same TF-IDF representation and balanced class weights.

Models were selected using validation macro F1-score because macro F1 gives equal importance to every class.

Selected sentiment model: **logistic_regression**

Selected topic model: **logistic_regression**

Final models were saved as:

```
round2/models/sentiment_model.pkl  
round2/models/topic_model.pkl
```

5. Training Methodology

Split	Records
Training	5382
Validation	1805
Testing	1813

A group-aware split used the original text as the group. Therefore identical texts could not cross dataset splits.

Train/validation overlap: 0

Train/test overlap: 0

Validation/test overlap: 0

The selected model was retrained on training plus validation records and evaluated once on the untouched test set.

6. Evaluation Metrics

Sentiment classification

Metric	Score
Accuracy	0.5885
Macro precision	0.5913
Macro recall	0.5887
Macro F1-score	0.5897
Weighted F1-score	0.5894

Per-class results:

Class	Precision	Recall	F1-score	Support
Negative	0.6481	0.6385	0.6433	603
Neutral	0.5138	0.5474	0.5301	612
Positive	0.6120	0.5803	0.5957	598

Topic classification

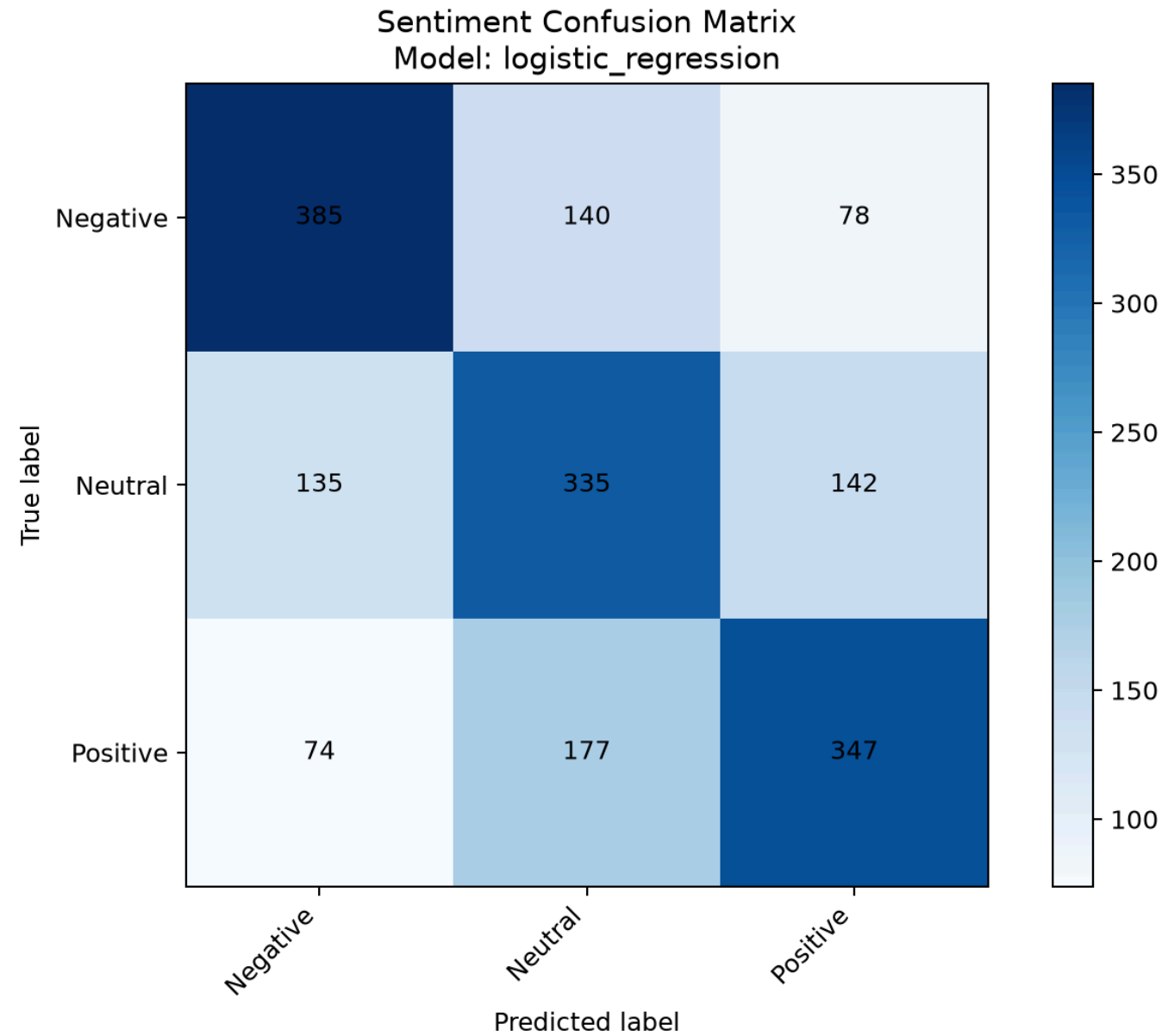
Metric	Score
Accuracy	0.9073
Macro precision	0.7890
Macro recall	0.5091
Macro F1-score	0.5875
Weighted F1-score	0.8948

Per-class results:

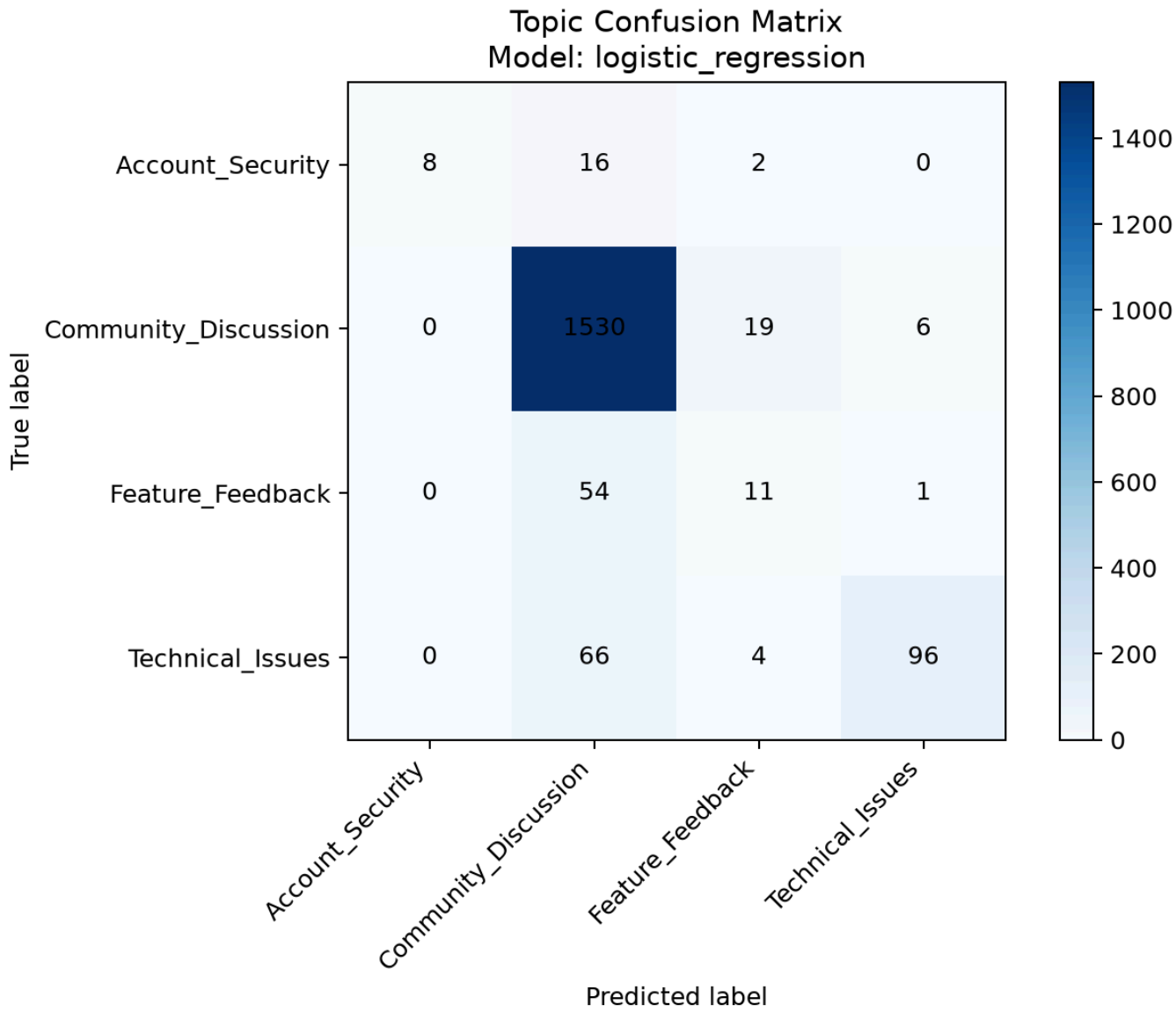
Class	Precision	Recall	F1-score	Support
Account_Security	1.0000	0.3077	0.4706	26
Community_Discussion	0.9184	0.9839	0.9500	1555
Feature_Feedback	0.3056	0.1667	0.2157	66
Technical_Issues	0.9320	0.5783	0.7138	166

7. Confusion Matrices

Sentiment



Topic



The matrices show strong majority-class performance and more difficulty with neutral sentiment and minority topics.

8. Error Analysis

The error files were generated from final test predictions:

```
round2/reports/sentiment_errors.csv
round2/reports/topic_errors.csv
```

Sentiment error examples

Text ID	Actual label	Predicted label	Text
TXT_00003	Positive	Neutral	Would you like to join us at our annual gala at the Sandman Signature Resort on Oct 25? Contact our centre at 6042773100 for tickets!

Text ID	Actual label	Predicted label	Text
TXT_00006	Negative	Positive	@user @user aaaah. Nokia used to make the Best Phone Cameras Ever. Sadly I think those days may be past.
TXT_00010	Neutral	Positive	"1st game played under Solar Powered floodlights in the world happened in MYSA,Nairobi Kenya. #PhilipsLED"
TXT_00033	Positive	Negative	The game 1st album 1 song he say he rap like Eazy-E then Nas then 50 cent then dr Dre then Jay-Z then Eazy-E again then like his dead bro
TXT_00108	Negative	Neutral	November 21 in the lonely hour tour.. OKAY SAM SMITH OKAY!!!! 😞(((TAPOS SOLD OUT PA OKAY!!!!!! OKAY LANG TALAGA

Typical causes include sarcasm, slang, mixed sentiment, short text, and neutral language containing emotional words.

Topic error examples

Text ID	Actual label	Predicted label	Text
TXT_00079	Account_Security	Community_Discussion	bloodymary - hackers nirvana - smell like teen spirit butterfingers - nicotine feeder - 7 days in the sun smashing pumpkins - bullets with..
TXT_00190	Feature_Feedback	Community_Discussion	What's the requirement to work at Dunkin? Complete up to 8th grade? Reasons I hate going there. No people skills.
TXT_00261	Technical_Issues	Feature_Feedback	"Kendrick Lamar may just be the greatest rapper of this generation, and maybe many more to come"
TXT_00296	Technical_Issues	Community_Discussion	SUN UPDATE: UFC 191, Beyonce with Ronda Rousey, Jimmy Snuka, TUF, New WWE stable, CHIKARA King of Trios
TXT_00402	Feature_Feedback	Community_Discussion	Fuck a Saturday class I just wanna go home and play with our squirrel and watch Jurassic Park

Topic errors are concentrated in minority categories because those categories contain fewer training examples and share vocabulary with other topics.

9. Limitations

The topic dataset is highly imbalanced. The model performs better on Community Discussion than on the smallest classes.

TF-IDF cannot fully understand sarcasm, outside context, world knowledge, or subtle meaning.

Future improvements could include character features, transformer embeddings, cross-validation, hyperparameter tuning, and additional minority-class data.

10. Reproducibility

Run these commands from the project root:

```
python "round2\src\inspect_dataset.py"  
python "round2\src\check_text_duplicates.py"  
python "round2\src\train_models.py"  
python "round2\src\generate_reports.py"
```

11. Conclusion

This project rebuilds the Social Engine semantic layer using text normalization, duplicate-aware splitting, TF-IDF features, model comparison, final test evaluation, confusion matrices, and error analysis.

The resulting system performs sentiment recognition and topic classification with transparent evaluation and documented limitations.